PYTHON LOGICAL PROGRAMMING ASSIGNMENT

Topics: List, Tuple, Set and Dictionary | 20 Practice Questions

Student Name: _____ **Date:** _____

SECTION 1: LIST-BASED QUESTIONS

1. Remove Duplicates: Write a Python program to remove duplicate elements from a list without using `set()`.

2. Second Largest: Write a program to find the second-largest element from a list without using `sort()`, `sorted()` or `max()`.

3. Separate Even and Odd: Write a program to separate even and odd numbers from a list into two different lists.

4. Element Frequency: Write a program to count how many times each element occurs in a list.

5. Move Zeros: Write a program to move all zeros to the end of a list without changing the order of the other elements.

SECTION 2: TUPLE-BASED QUESTIONS

6. Tuple Reversal: Write a program to reverse a tuple without using slicing.

7. Maximum and Minimum: Write a program to find the maximum and minimum elements from a tuple without using built-in functions.

8. Tuple Element Frequency: Write a program to count the occurrence of a particular element in a tuple without using the `count()` method.

9. Convert Tuple: Write a program to convert a tuple into a list, remove duplicate elements, and convert it back into a tuple.

10. Nested Tuple: Consider the following tuple and write a program to find the student who scored the highest marks.

```
students = (("Amit", 85), ("Rahul", 92), ("Sneha", 78), ("Priya", 95))
```

SECTION 3: SET-BASED QUESTIONS

11. Common Elements: Write a program to find common elements between two lists using sets.

12. Unique Elements: Write a program to find all unique elements from two lists combined.

13. Subset Checking: Write a program to check whether one set is a subset of another set without using the `issubset()` method.

14. Remove Common Elements: Write a program to remove all common elements between two sets.

15. Symmetric Difference: Write a program to find elements present in either of two sets but not in both, without using the `symmetric_difference()` method.

SECTION 4: DICTIONARY-BASED QUESTIONS

16. Character Frequency: Write a program to count the frequency of each character in a given string using a dictionary. Example: `"programming"`.

17. Word Frequency: Write a program to count the frequency of every word in a sentence using a dictionary. Example: `"python is easy and python is powerful"`.

18. Highest Marks: Consider the dictionary below and write a program to find the student with the highest marks without using `max()`.

```
students = {"Amit": 85, "Rahul": 92, "Sneha": 78, "Priya": 95}
```

19. Merge Dictionaries: Write a program to merge two dictionaries. If a key exists in both dictionaries, add their values.

```
d1 = {"a": 10, "b": 20}
d2 = {"b": 30, "c": 40}
```

20. Sort Dictionary by Values: Write a program to arrange dictionary elements in ascending order based on their values without using `sorted()`.

ASSIGNMENT INSTRUCTIONS

- Use loops and conditional statements wherever required.
- Do not use shortcut functions such as max(), min(), sort() and sorted() unless the question allows them.
- Use collection methods wherever appropriate.
- Do not hardcode the final output.
- Test each program with different input values and be prepared to explain the logic.

Submission: Submit all 20 programs with their outputs as instructed by your trainer.